
Education

2024- **Gatsby Computational Neuroscience Unit, University College London**

PhD in Computational Neuroscience and Machine Learning

Supervisor: Maneesh Sahani. Working on probabilistic approaches to identify latent non-linear dynamical systems (incl. from neural population activity) using approximate inference and Gaussian processes.

2023–2024 **University of Cambridge**

MPhil Machine Learning and Machine Intelligence, Distinction

Modules: Introduction to Machine Learning, Probabilistic Machine Learning, Deep Learning and Structured Data, Ethical Considerations in Machine Learning and Machine Intelligence, Speech Recognition, Advanced Machine Learning, Reinforcement Learning and Decision Making, Geometric Deep Learning, Computational Neuroscience.

Awards:

- Google DeepMind Cambridge Scholarship.

2022–2023 **University College London**

MSc Neuroscience, Distinction

Modules: Receptors and Synaptic Signalling, Developmental Neurobiology, Systems and Circuits Neuroscience, Cognitive Systems Neuroscience.

Awards:

- Best Overall Student Prize: 'Highest overall average mark from the 2022/23 cohort.' November 2023.
- The Dean's List - Faculty of Life Sciences.

2019–2022 **University College London**

BSc Psychology, First Class Honours

Chosen modules for Year 3: Topics in Neurobiology, Computer Programming, Advanced Multivariate Statistical Methods, Cognitive Neuroscience, Applied Decision-making, Social Psychology.

Awards:

- Cecily De Monchaux Memorial Prize: 'Most Outstanding Third Year Student.' July 2022.
- Faculty Medal and Prize: 'Best Undergraduate Student Across the UCL Faculty of Brain Sciences.' July 2022.
- The Dean's List - Faculty of Brain Sciences: 'A commendation to undergraduate students excelling in their chosen field'. July 2022.
- First Year Sessional Prize A: 'Recognition of being one of the two students with the best exam performance in Year 1 during the 2019-20 academic session.' July 2020.

Research Experience

2026 **Princeton Reinforcement Learning (RL) Lab, Princeton University**

- Visiting Student Research Collaborator (VSRC) advised by Benjamin Eysenbach.

2024 **Computational and Biological Learning Lab (CBL), University of Cambridge**

- MPhil Research Project, supervised by Guillaume Hennequin.
- Nonlinear, Scalable Sequential Component Analysis (non-linear dimensionality reduction method to extract temporally non-reversible neural manifolds).

2023 **Gatsby Computational Neuroscience Unit, University College London**

- MSc Research Project, supervised by Maneesh Sahani and Jennifer Linden.
- An Autoencoder-based Model of Biological Auditory Representation Learning.

2021–2022 **MetaLab, University College London**

- BSc Research Project, supervised by Nadine Dijkstra and Stephen Fleming.
- Psychophysical investigation of the mechanisms underlying perceptual source confusion (arising when visual mental imagery is wrongfully attributed to visual perception, or vice versa, due to the overlap in the neural codes they activate).

2021 **Eccentric Vision Lab**, *University College London*

- Research Assistant, supervised by John Greenwood.
- Pre-processing and statistically analysing participant data using Matlab for different research projects pertaining to human visual perception.

Publications and Conference Posters

- L. Fox, K. Biegún, J. Heald, S. Hromádka, **A. Rosinski**, M. Sahani. RAMP: Recognition parametrisation by Amortised Message Passing. *ProbML*, 2026.
- **A. Rosinski**, V. Ruetten., G. Hennequin. Identifying Neural Activity Manifolds through Non-reversibility Analysis. *Cosyne 2026*.
- L. Fox*, **A. Rosinski***, J. Linden., M. Sahani. Investigating the computational principles underlying dynamical auditory cortical representations. *Cosyne 2026*.
- L. Fox*, **A. Rosinski***, J. Linden., M. Sahani. Investigating the computational principles underlying auditory cortical representations using recurrent networks. *International Conference on Auditory Cortex (ICAC) 2025*.
- P. Monteagudo-Lago*, **A. Rosinski***, A. J. Dudzik, P. Veličković. Asynchrony Invariance Loss Functions for Graph Neural Networks. *Proceedings of the Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) at the 41 st International Conference on Machine Learning, Vienna, Austria. PMLR 251, 2024*.
- C. Slinn, Z. Nikodemova, **A. Rosinski**, Dijkstra, N. Vividness of visual imagery predicts performance on a visual working memory task when an imagery strategy is encouraged. *PsyArXiv*, 2023.

Teaching

- Teaching ODEs for the Gatsby Bridging Programme (2026).
- Teaching assistant for the Gatsby Unit Courses: Probabilistic and Unsupervised Learning (2025/26), Approximate Inference and Learning in Probabilistic Models (2025/26).

Technical Skills

- Programming: Python (incl. NumPy, PyTorch, Jax, Scikit-Learn) + experience with Matlab and R.